

**MATH 220, Matrices**  
**Penn State University, University Park**  
**Spring 2026**

## **COURSE INFORMATION**

**CATALOG DESCRIPTION: MATH 220 (GQ) MATRICES (3 credits)** Systems of linear equations appear everywhere in mathematics and its applications. MATH 220 will give students the basic tools necessary to analyze and understand such systems. The initial portion of the course teaches the fundamentals of solving linear systems. This requires the language and notation of matrices and fundamental techniques for working with matrices such as row and column operations, echelon form, and invertibility. The determinant of a matrix is also introduced; it gives a test for invertibility. In the second part of the course the key ideas of eigenvector and eigenvalue are developed. These allow one to analyze a complicated matrix problem into simpler components and appear in many disguises in physical problems. The course also introduces the concept of a vector space, a crucial element in future linear algebra courses.

**PREREQUISITE:** MATH 110 or MATH 140 or MATH 140B or MATH 140E or MATH 140G or MATH 140H

**TEXTBOOK:** MyLab Math with Pearson eText -- Instant Access -- for Linear Algebra and its Applications (18-Weeks) ISBN-13: 9780135851203

**COURSE FORMAT:** There are three 50-minute lectures each week. The sections covered in class are listed at the end of this syllabus.

### **IMPORTANT COURSE DATES:**

- **Midterm Examinations: Wednesday, February 18 and Wednesday, April 1**
- **Makeup Midterm Examinations: Thursday, February 26 and Thursday, April 9**
- **Final Examination: During the week of finals (May 4-8)**

See the Assessments section below for more information on examinations.

## **COURSE GOALS AND OBJECTIVES**

### **COURSE GOALS**

1. Analyze and solve a system of linear equations.
2. Understand the important characteristics of matrices such as its fundamental subspaces, rank, determinant, eigenvalues, and eigenvectors.
3. Learn how to use characteristics of a matrix to solve a system of linear equations or study properties of a linear transformation.
4. Understand important concepts of vector spaces such as independence, basis, dimension and orthogonality.
5. Use linear algebra terminology to express problems in branches of science; communicate solutions back in terms of the original context.
6. Apply linear algebra to computationally robust problems using Mathematica or another computing program.

### **COURSE OBJECTIVES**

#### Section 1.1 – Systems of Linear Equations

1. Use elementary row operations to solve systems of linear equations.
2. Determine if a system of linear equations is consistent.
3. Determine the conditions for which a linear system is consistent.
4. Determine the validity of statements about systems of linear equations, row operations, or matrices.

#### Section 1.2 – Row Reduction and Echelon Forms

1. Identify matrices in echelon form and reduced echelon form.
2. Row reduce matrices to reduced echelon form.
3. Find the general solution to a system with a given augmented matrix.

4. Determine if a solution is consistent given a description of the corresponding coefficient matrix.
5. Determine the conditions for which a linear system has specified types of solutions.
6. Determine the validity of statements about row reduction and echelon forms.
7. Find interpolating polynomials for small data sets using row reduction.
8. Solve applications using row reduction.

### Section 1.3 – Vector Equations

1. Compute sums and scalar products of vectors, both algebraically and geometrically.
2. Convert between vector equations and systems of equations.
3. Determine if a vector is a linear combination of other vectors.
4. Characterize the span of a set of vectors algebraically or geometrically.
5. Determine the validity of statements about vectors and vector equations.
6. Solve applications involving vector equations.

### Section 1.4 – The Matrix Equation $Ax=b$

1. Compute the product of a matrix and a vector.
2. Convert between matrix equations, vector equations, and systems of equations.
3. Solve matrix equations using augmented matrices.
4. Characterize the span of the column vectors of a matrix.
5. Determine whether a matrix equation has no solution, one solution, or infinitely many solutions.
6. Determine the validity of statements about vector equations and matrix equations.
7. Solve applications involving matrix equations.

### Section 1.5 – Solution Sets of Linear Systems

1. Determine if a system of equations has a nontrivial solution.
2. Solve a system of equations or a matrix equation and write the solution in parametric form.
3. Describe the solution sets of systems of equations geometrically.
4. Determine the validity of statements about solution sets of linear equations.

### Section 1.6 – Applications of Linear Systems

1. Setup and solve linear difference equations representing changes in population.
2. Describe and solve network flow diagrams using linear systems.
3. Use vector equations to balance chemical equations.
4. Interpret solutions to applications involving linear systems.

### Section 1.7 – Linear Independence

1. Determine if a set of vectors is linearly independent and determine if a vector is in a given span.
2. Determine conditions for which vectors are linearly independent or have a given span.
3. Determine the validity of statements about linear independence.

### Section 1.8 – Introduction to Linear Transformations

1. Algebraically find the image of a given vector under a linear transformation.
2. Given a linear transformation  $T(x)=Ax$ , find  $x$  for a given  $b$  in the image of  $T$ .
3. Determine the conditions for which a linear transformation has a given domain and codomain.
4. Determine if a vector is in the range of a linear transformation.
5. Geometrically describe the image of a vector under a linear transformation.
6. Use the linearity of transformations to find the images of vectors under the transformation.
7. Determine the validity of statements about linear transformations.

### Section 1.9 – The Matrix of a Linear Transformation

1. Find the standard matrix of a linear transformation.
2. Find vectors whose images under a linear transformation are given.
3. Determine the validity of statements about properties of linear transformations.
4. Determine if linear transformations are one-to-one or onto.

### Section 2.1 – Matrix Operations

1. Compute sums, products, and scalar products of matrices.
2. Find values of matrices such that products of matrices have given properties.
3. Determine the validity of statements about matrix operations.

### Section 2.2 – The Inverse of a Matrix

1. Find the inverse of a  $2 \times 2$  matrix using the formula.
2. Use the inverse of a matrix to solve a linear system.
3. Determine the validity of statements about inverses of matrices.
4. Solve equations involving invertible matrices.
5. Find the inverse of a matrix using row reduction.

### Section 2.3 – Characterizations of Invertible Matrices

1. Use the Invertible Matrix Theorem to determine the validity of statements about matrices.
2. Find the inverse of a linear transformation.

### Section 3.1 – Introduction to Determinants

1. Compute determinants using cofactor expansions.
2. Determine the effect of elementary row operations on a determinant.
3. Calculate determinants of scalar multiples of matrices.

### Section 3.2 – Properties of Determinants

1. Identify properties of determinants
2. Find determinants by row reduction to echelon forms.
3. Use properties of determinants to evaluate determinants.
4. Prove properties of determinants.
5. Use determinants to determine if a matrix is invertible or a set of vectors is linearly independent.
6. Determine the validity of statements about properties of determinants.

### Section 4.1 – Vector Spaces and Subspaces

1. Determine whether a given set is a vector space.
2. Determine whether a given set is a subspace.
3. Determine the validity of statements about vector spaces and subspaces.
4. Solve applications involving vector spaces and subspaces.

### Section 4.2 – Null Spaces, Column Spaces, and Linear Transformations

1. Determine whether a vector is in the null or column space of a matrix.
2. Find a nonzero vector in the null, column, or row space of a matrix.
3. Determine the validity of statements about linear transformations and null, column, and row spaces.
4. Use the theory of null and column spaces to find solutions of linear systems.

### Section 4.3 – Linearly Independent Sets, Bases

1. Determine whether a set is linearly independent, or is a basis for, or spans a vector space.
2. Find bases for the null, column, or row space of a matrix.
3. Find a basis for the span of a set of vectors.
4. Determine the validity of statements about linearly independent sets and bases.

### Section 4.5 – The Dimension of a Vector Space

1. Find the dimension of a subspace.
2. Find the dimensions of the null, column, and row spaces for a matrix.

### Section 5.1 – Eigenvectors and Eigenvalues

1. Determine if a vector or number is an eigenvector or eigenvalue of a given matrix.
2. Find a basis for the eigenspace corresponding to an eigenvalue.
3. Find the eigenvalues of matrices.
4. Determine the validity of statements about eigenvectors and eigenvalues.

### Section 5.2 – The Characteristic Equation

1. Find the characteristic polynomial and eigenvalues of a  $2 \times 2$  matrix.
2. Find the characteristic polynomial of a  $3 \times 3$  matrix.
3. Find the eigenvalues of triangular matrices.

### Section 5.3 – Diagonalization

1. Compute  $A^k$  for  $A = PDP^{-1}$ .
2. Use the diagonalization theorem to find the eigenvalues of a matrix and a basis for each eigenspace.

3. Diagonalize matrices.
4. Determine the validity of statements about diagonalization.
5. Determine whether matrices are diagonalizable.
6. Answer conceptual questions about diagonalization of matrices.

#### Section 6.1 – Inner Product, Length and Orthogonality

1. Perform operations on vectors using inner products.
2. Find a unit vector.
3. Find the distance between two vectors.
4. Determine whether two vectors are orthogonal.

#### Section 6.2 – Orthogonal Sets

1. Determine whether a set of vectors is orthogonal.
2. Find an orthogonal projection onto a line through a given vector and the origin.
3. Write a vector as a sum of two orthogonal vectors.
4. Find the distance between a vector and a line through the origin.
5. Determine whether a set of vectors is orthonormal.
6. Determine the validity of statements about orthogonal sets or matrices and orthogonal projections.

#### Section 6.3 – Orthogonal projections

1. Find the orthogonal decomposition of a vector.
2. Find an orthogonal projection onto a given subspace.
3. Use the best approximation theorem to find the closest point or a distance.
4. Construct a vector that is orthogonal to a given orthogonal set.
5. Determine the validity of statements about orthogonal projections onto subspaces.

#### Section 6.4 – The Gram-Schmidt Process

1. Use the Gram-Schmidt process to produce an orthogonal basis and find an orthonormal basis.

#### Section 6.5 – Least-Squares Problems

1. Use normal equations to find least-squares solutions of systems.
2. Determine the validity of statements about least-squares solutions.

#### Section 7.1 – Diagonalization of Symmetric Matrices

1. Determine the validity of statements about symmetric matrices.
2. Use the Spectral Theorem for Symmetric Matrices to find a spectral decomposition of symmetric matrices.

#### Section 7.4 – The Singular Value Decomposition

1. Find the singular values of a matrix.
2. Find a singular value decomposition of a matrix.
3. Use the singular value decomposition of a matrix to find particular values or vectors.

This course satisfies the General Education learning objectives Key Literacies and Critical & Analytical Thinking.

## **GETTING HELP OUTSIDE OF CLASS**

**OFFICE HOURS:** Weekly office hours are available. See Canvas for a schedule of hours.

**TUTORS and PENN STATE LEARNING:** Free mathematics tutoring is available at Penn State Learning. You can find information [here](#) about their online and in-person options.

**LA SESSIONS:** Each week learning assistants (LAs), who are part of the instructional team, will hold multiple evening problem sessions. These sessions are a continuation of what is discussed in class each week. By attending these sessions, you are strengthening your skills and understanding of the content. More information about the general structure, time, and location of sessions will be communicated via Canvas.

## ASSESSMENTS

**MIDTERM EXAMINATIONS:** Two 75-minute evening examinations will be given during the semester and a 110-minute comprehensive final exam will be given during the final exam period. No books, notes, or calculators may be used on the examinations. You must bring your University ID card to all exams. The midterm examinations will be given from **6:15 to 7:30 p.m.** on the following dates:

|                  |                        |
|------------------|------------------------|
| <b>Midterm 1</b> | Wednesday, February 18 |
| <b>Midterm 2</b> | Wednesday, April 1     |

Rooms for examinations will be announced by your instructor at a later date. It is not permissible to take the exam in a different instructor's assigned room.

**FINAL EXAMINATION:** The final examination will be given during finals week, May 4-8. **The final examination may be scheduled on any day during the final examination period. Do not plan to leave University Park until after Friday, May 8.** Students may access their final exam schedules on Monday, February 16, through their LionPATH account. Notification of conflicts is given on the student's final exam schedule. Final exam conflicts must be submitted through Lionpath by Sunday, March 8.

**HOMEWORK ASSIGNMENTS:** There are two types of homework assignments. Written homework problem sets will be assigned by your instructor. Credit is given for writing full and complete solutions to the problems assigned. A scanned copy of the completed work must be submitted to your instructor through your Gradescope account.

Online homework assignments are completed through your MyLab Math account. You are responsible for activating your MyLab Math account no later than Friday, January 16.

**LATE HOMEWORK ASSIGNMENT POLICY:** Due dates for every assignment are provided on Canvas. Unless otherwise stated, assignments are due on those dates. Students must notify their instructor via email with a valid excuse within 24 hours of an assignment due date to request an extension. The instructor will review late requests and circumstances on a case-by-case basis and make decisions accordingly. Late assignments will only be accepted with prior permission.

**QUESTIONS REGARDING ASSIGNMENT GRADES:** Questions regarding the grading of homework assignments should be directed to your section instructor within two weeks of the date the assignment is graded. Questions regarding the grading of exams or missing exam grades should be directed to your section instructor as soon as possible. **Students and instructors will have until the next exam for the course takes place to review a physical copy of the exam or report a missing grade.** Students will have one week from the date of the final exam to contact their instructor regarding questions about the final exam or a missing final exam grade.

## AUTHORIZED EARLY EXAM AND MAKEUP EXAM POLICIES

**AUTHORIZED EARLY EXAM AND MAKEUP EXAM POLICY:** In addition to the regularly scheduled midterm examinations, the math department schedules two additional options: An authorized early exam for each of the midterms from 4:50 – 6:05 p.m. on the same night as the regularly scheduled exam and a makeup exam scheduled on an evening different from that of the regularly scheduled exam. Students who attend the authorized early exam will not be permitted to leave before 6:10 p.m. A valid reason is required to sign up for either of these exams; when signing up, you must include a computer printout of your academic schedule, including your name, so your instructor can validate your request. It is the student's responsibility to sign up and to note the time and location of the makeup or conflict exam.

**Who may take the Authorized Early Exam?** If a student has a valid, documented conflict with the regular examination time, such as a class or another official universal activity, they may sign up for the authorized early exam. The deadline to sign up for an authorized early exam for midterm 1 is Monday, February 16 and the deadline for midterm 2 is Monday, March 30. If a student has not signed up for the authorized early exam by these deadlines, they will not be permitted to take the exam.

**Instructions on Authorized Early Exam night.** The student is responsible for knowing the room and time of the authorized early examination. Each student must bring their University ID to the authorized early examination. The ID will be checked by the proctor. Although the authorized early examination will end at 6:05 p.m., no student will be permitted to turn on their cell phone nor leave the examination room before 6:10 p.m. Any student who leaves before 6:10 p.m. will receive a grade of zero on the examination and will not be allowed to retake it.

**Who may take the Makeup Exam?** Students who have a valid, documented reason, such as a class conflict, during both the authorized early examination and regular examination times are permitted to schedule a makeup examination with no penalty. The student must be prepared to verify the reason for taking the makeup. **Personal business, such as travel, employment, weddings, graduations, or attendance at public events such, as concerts, sporting events, and Greek Rush events, is not a valid excuse. Forgetting the date, time, or room of an examination is not a valid excuse.** Students who do not have a valid reason for missing the examination may be permitted to schedule the makeup, but 20% will be deducted from their score. Students who have taken either the regularly scheduled examination or authorized early examination are not permitted to take the makeup examination. The makeup examination will be given from **6:15 – 7:30 p.m.** on the following dates:

|                         |                       |
|-------------------------|-----------------------|
| <b>Makeup Midterm 1</b> | Thursday, February 26 |
| <b>Makeup Midterm 2</b> | Thursday, April 9     |

**How and when to sign up for the Makeup Exam.** A student who is ill on exam night must contact their instructor within 24 hours of the exam. Students must sign up for the Makeup Exam as soon as possible following the regular exam date. The student is responsible for knowing the room and time of the makeup examination. If a student has not signed up with their instructor, the student will not be allowed to take the makeup exam. The deadline to sign up for a makeup exam for midterm 1 is Monday, February 16 and the deadline for midterm 2 is Monday, March 30.

**Instructions on Makeup Exam night.** The student is responsible for knowing the room and time of the makeup examination. Each student must bring their PSU ID to the makeup examination. The ID will be checked by the proctor.

**What if a student misses both the regularly scheduled exam and the makeup exam?** If a student misses both the regularly scheduled examination and the scheduled makeup due to a valid, verifiable reason, it may be possible to take the makeup examination by appointment. All such makeup examinations must be scheduled through the classroom instructor with the approval of the course coordinator and must be completed no later than one week after being able to return to classwork. Only one makeup opportunity per exam is permitted. Barring emergencies, makeup examinations cannot be rescheduled. Students who do not have a valid, verifiable reason for missing an exam or who do not complete the makeup exam within one week will receive a grade of zero on the exam.

**CONFLICT FINAL EXAM POLICY** There are two types of conflict final examinations: direct and overload. Direct conflicts are two examinations scheduled at the same time. Overload examinations are defined as three or more examinations scheduled in consecutive time periods or within one calendar day. Students may elect to take the three or more examinations on the same day if they wish or request a conflict final examination. **A student must take action to request a conflict exam through LionPATH between Monday, February 16 and Sunday, March 8.** **Conflict final examinations cannot be scheduled through the Mathematics department.** Students who miss or cannot take the final examination due to a valid and documented reason, such as illness, may be allowed to take a makeup final examination at the beginning of the next semester. **Personal business, such as travel, employment, weddings, graduations, or attendance at public events such, as concerts and sporting events is not a valid excuse. Forgetting the date, time, or room of an examination is not a valid excuse.** If the student does not have a valid reason, as explained above, a 20% penalty will be imposed. All such makeup examinations must be arranged through the instructor with the approval of the course coordinator, and students in such a situation should contact their instructors within 24 hours of the scheduled final examination. Students who have taken the original final examination are not permitted to take a makeup final examination.

## GRADING POLICY

**COURSE GRADES:** Grades will be assigned out of 100 percent, distributed as follows:

|                          |      |
|--------------------------|------|
| Midterm Exam 1           | 20%  |
| Midterm Exam 2           | 20%  |
| Final Exam               | 25%  |
| Quizzes and Section Work | 8%   |
| MyLab Math Homework      | 12%  |
| Written Homework         | 15%  |
| Total                    | 100% |

Final course grades will be assigned as follows:

| Grade     | Percent Score |
|-----------|---------------|
| <b>A</b>  | 93% - 100%    |
| <b>A-</b> | 90%-92%       |
| <b>B+</b> | 87%-89%       |
| <b>B</b>  | 83%-86%       |
| <b>B-</b> | 80%-82%       |
| <b>C+</b> | 77%-79%       |
| <b>C</b>  | 70%-76%       |
| <b>D</b>  | 60%-69%       |
| <b>F</b>  | 0%-59%        |

The unavoidable consequence is that some students will be “just a point” away from the next higher or lower grade. For reasons of fairness, the policy in this course is to **NOT** adjust individual grades in such circumstances. **NOTE:** Each student's letter grade will be based **EXCLUSIVELY** on the examinations, homework, quizzes, and section work. **There is no "extra-credit" work.**

**DEFERRED GRADES:** Students who are **currently passing a course** but are unable to complete the course because of illness or an emergency may be granted a deferred grade which will allow the student to complete the course within ten weeks after the final grade reporting deadline for the semester. Note that deferred grades are limited to those students who can verify and document a valid reason for not being able to take the final examination. For more information see [Deferred Grades](#).

## GENERAL UNIVERSITY POLICIES

**LATE-DROP:** Students may add/drop a course without academic penalty within the first six calendar days of the semester. A student may late drop a course within the first twelve weeks of the semester. After the first six days and before deciding to late drop this course, each student should consult with their academic advisor. The late drop deadline is **Friday, April 10**.

**ACADEMIC INTEGRITY:** According to Penn State policy G-9: Academic Integrity, an academic integrity violation is “an intentional, unintentional, or attempted violation of course or assessment policies to gain an academic advantage or to advantage or disadvantage another student academically.” Unless your instructor tells you otherwise, you must complete all course work entirely on your own, using only sources that have been permitted by your instructor, and you may not assist other students with papers, quizzes, exams, or other assessments. If your instructor allows you to use ideas, images, or word phrases created by another person (e.g., from Course Hero or Chegg) or by generative technology, such as ChatGPT, you must identify their source. You may not submit false or fabricated information, use the same academic work for credit in multiple courses, or share instructional content. Students with questions about academic integrity should ask their instructor **before submitting work**. **Students facing allegations of academic misconduct may not drop/withdraw from the affected course** unless they are cleared of wrongdoing (see [G-9: Academic Integrity](#)). Attempted drops will be prevented or reversed, and students



will be expected to complete course work and meet course deadlines. Students who are found responsible for academic integrity violations face academic outcomes, which can be severe, and put themselves at jeopardy for other outcomes which may include ineligibility for Dean's List, pass/fail elections, and grade forgiveness. Students may also face consequences from their home/major program and/or The Schreyer Honors College.

In order to ensure that all students have a fair and equal opportunity to succeed in this course, the Mathematics Department is committed to enforcing the University's academic integrity policy. Below is a description of academic misconduct and the department's responsibilities when misconduct is suspected.

### **Academic Misconduct**

In this course, academic misconduct includes, but is not limited to:

- Copying the work of another student on an exam, quiz, or assignment;
- Passing off the work of another individual as your own;
- Using non-approved devices or aids, such as generative AI, on exams, quizzes, or assignments;
- Having unauthorized possession of exams or quizzes;
- Engaging in deception in order to extend or reschedule an exam, quiz, or assignment;
- Facilitating acts of academic misconduct by others.

### **When Academic Misconduct is Suspected**

If a student is suspected of academic misconduct, the instructor's duties are to:

- Confidentially inform the student of the allegation;
- Enter the charge and recommended sanctions on an Eberly College of Science Academic Integrity form;
- Proceed with the Academic Integrity charge process. The student can choose to accept or contest the allegation at this point.

Note that a student's refusal to respond to the charges within a reasonable period of time is construed as acceptance of the allegation and proposed sanctions. Once the Academic Integrity form has been accepted or contested by the student, it is sent to the College's Academic Integrity Committee for adjudication. A student cannot drop or withdraw from the course during the adjudication process.

### **Sanctions**

If a student accepts an academic misconduct allegation, or if they are found guilty during adjudication, probable sanctions include:

- A warning and
- Reduction of the assignment grade to zero or
- Reduction of the quiz or exam grade to zero.

Additional sanctions might include:

- Reduction in the final course grade;
- An F in the course.

In addition, the student will be unable to drop or withdraw from the course.

Please see the [Eberly College of Science Academic Integrity homepage](#) for additional information and procedures.

**STUDENTS WITH DISABILITIES:** Penn State welcomes students with disabilities into the University's educational programs. If you have a disability-related need for reasonable academic adjustments in this course, contact Student Disability Resources at 814-863-1807 (V/TTY). For further information, please visit Student Disability Resources web site: <http://equity.psu.edu/student-disability-resources/>.

In order to receive consideration for accommodations, you must contact SDR and provide documentation (see the documentation guidelines at <http://equity.psu.edu/student-disability-resources/>). If the documentation supports your request for reasonable accommodations, SDR will provide you with an accommodation letter identifying appropriate academic adjustments. Please share this letter with your instructors and discuss the accommodations with them as early in your courses as possible. You must follow this process for every semester that you request accommodations.

**CODE OF MUTUAL RESPECT AND COOPERATION:** The Eberly College of Science Code of Mutual Respect and Cooperation pertains to all members of the college community; faculty, staff, and students. The [Code of Mutual Respect and Cooperation](#) was developed to embody the values that we hope our faculty, staff, and students possess, consistent with the aspirational goals expressed in the Penn State Principles. The University is strongly committed to freedom of expression, and consequently, the Code does not constitute University or College policy,



and is not intended to interfere in any way with an individual's academic or personal freedoms. We hope, however, that individuals will voluntarily endorse the 12 principles set forth in the Code, thereby helping us make the Eberly College of Science a place where every individual feels respected and valued, as well as challenged and rewarded.

**EDUCATIONAL EQUITY:** The Office of the Vice Provost for Educational Equity serves as a catalyst and advocate for Penn State's diversity and inclusion initiatives. Educational Equity's vision is a Penn State community that is an inclusive and welcoming environment for all. If you wish to learn more or if you wish to [report bias](#), please visit the Educational Equity [website](#).

**COUNSELING AND PSYCHOLOGICAL SERVICES:** Many students at Penn State face personal challenges or have psychological needs that may interfere with their academic progress, social development, or emotional wellbeing. The university offers a variety of confidential services to help you through difficult times, including individual and group counseling, crisis intervention, consultations, online chats, and mental health screenings. These services are provided by staff who welcome all students and embrace a philosophy respectful of clients' cultural and religious backgrounds, and sensitive to differences in race, ability, gender identity and sexual orientation.

[Counseling and Psychological Services at University Park \(CAPS\)](#)

Penn State Crisis Line (24 hours/7 days/week): 877-229-6400

Crisis Text Line (24 hours/7 days/week): Text LIONS to 741741

**QUESTIONS, PROBLEMS, OR COMMENTS:** If you have questions or concerns about the course, please consult your instructor first. If further guidance is needed, you may contact the course coordinator whose contact information is given below.

Nathan Fontes, Lecturer  
421 McAllister Building  
University Park, PA 16802  
Email: [math220coord@psu.edu](mailto:math220coord@psu.edu)

Include your Name, Student ID, Course, and Section Number in any correspondence and always use your @psu.edu email address.

Math 220: Matrices - Spring 2026

|    | MONDAY                                                         |  | TUESDAY                  |  | WEDNESDAY                                                                    |  | THURSDAY                 |  | FRIDAY                                                                                |  |
|----|----------------------------------------------------------------|--|--------------------------|--|------------------------------------------------------------------------------|--|--------------------------|--|---------------------------------------------------------------------------------------|--|
| 1  | 12-Jan-26                                                      |  | 13-Jan-26                |  | 14-Jan-26                                                                    |  | 15-Jan-26                |  | 16-Jan-26                                                                             |  |
|    | Course Introduction<br>Course Documents, Section 1.1           |  |                          |  | Systems of Linear Equations<br>Section 1.1<br>Online HW0                     |  |                          |  | Elementary Row Operations<br>Section 1.1                                              |  |
| 2  | 19-Jan-26                                                      |  | 20-Jan-26                |  | 21-Jan-26                                                                    |  | 22-Jan-26                |  | 23-Jan-26                                                                             |  |
|    | MLK Jr. Day<br>No Class                                        |  |                          |  | Gauss-Jordan Elimination<br>Section 1.2<br>Written HW0, Online HW1           |  |                          |  | Solution Sets of Linear Systems<br>Section 1.2                                        |  |
| 3  | 26-Jan-26                                                      |  | 27-Jan-26                |  | 28-Jan-26                                                                    |  | 29-Jan-26                |  | 30-Jan-26                                                                             |  |
|    | Introduction to Technology<br>No Reading<br>Online HW2         |  |                          |  | Vector Equations<br>Section 1.3<br>Written HW1                               |  |                          |  | Linear Independence and Span<br>Section 1.7                                           |  |
| 4  | 02-Feb-26                                                      |  | 03-Feb-26                |  | 04-Feb-26                                                                    |  | 05-Feb-26                |  | 06-Feb-26                                                                             |  |
|    | The Matrix Equation $Ax=b$<br>Section 1.4<br>Online HW3        |  |                          |  | Population Models<br>Sections 1.6, 1.10<br>Written HW2                       |  |                          |  | Homogeneous and Non-Homog. Systems<br>Section 1.5                                     |  |
| 5  | 09-Feb-26                                                      |  | 10-Feb-26                |  | 11-Feb-26                                                                    |  | 12-Feb-26                |  | 13-Feb-26                                                                             |  |
|    | Intro to Linear Transformations<br>Section 1.8<br>Online HW4   |  |                          |  | Matrix of a Linear Transformation<br>Section 1.9<br>Written HW3              |  |                          |  | Linear Transformations<br>Section 1.9<br>End of Midterm 1 Content                     |  |
| 6  | 16-Feb-26                                                      |  | 17-Feb-26                |  | 18-Feb-26                                                                    |  | 19-Feb-26                |  | 20-Feb-26                                                                             |  |
|    | Matrix Operations<br>Section 2.1<br>Online HW5, Written HW4    |  |                          |  | Review Day<br>No Lecture<br>Exam 1 (Evening)                                 |  |                          |  | Matrix Operations<br>Section 2.1                                                      |  |
| 7  | 23-Feb-26                                                      |  | 24-Feb-26                |  | 25-Feb-26                                                                    |  | 26-Feb-26                |  | 27-Feb-26                                                                             |  |
|    | The Inverse of a Matrix<br>Section 2.2<br>Online HW6           |  |                          |  | Characterizations of Invertible Matrices<br>Section 2.3                      |  |                          |  | Introduction to Determinants<br>Section 3.1                                           |  |
| 8  | 02-Mar-26                                                      |  | 03-Mar-26                |  | 04-Mar-26                                                                    |  | 05-Mar-26                |  | 06-Mar-26                                                                             |  |
|    | Properties of Determinants<br>Section 3.2<br>Online HW7        |  |                          |  | Determinants Activity<br>Section 3.1, 3.2<br>Written HW5                     |  |                          |  | Vector Spaces and Subspaces<br>Section 4.1                                            |  |
|    | 09-Mar-26                                                      |  | 10-Mar-26                |  | 11-Mar-26                                                                    |  | 12-Mar-26                |  | 13-Mar-26                                                                             |  |
|    | Spring Break<br>No Class                                       |  | Spring Break<br>No Class |  | Spring Break<br>No Class                                                     |  | Spring Break<br>No Class |  | Spring Break<br>No Class                                                              |  |
| 9  | 16-Mar-26                                                      |  | 17-Mar-26                |  | 18-Mar-26                                                                    |  | 19-Mar-26                |  | 20-Mar-26                                                                             |  |
|    | Bases and Dimension<br>Section 4.3<br>Online HW8               |  |                          |  | Null and Column Spaces<br>Section 4.2<br>Written HW6                         |  |                          |  | Vector Spaces Activity<br>Sections 4.1, 4.2, 4.3                                      |  |
| 10 | 23-Mar-26                                                      |  | 24-Mar-26                |  | 25-Mar-26                                                                    |  | 26-Mar-26                |  | 27-Mar-26                                                                             |  |
|    | Eigenvectors and Eigenvalues<br>Section 5.1<br>Online HW9      |  |                          |  | Eigenvalues and Eigenvectors<br>Section 5.2<br>Written HW7                   |  |                          |  | Eigenvectors and Eigenvalues Activity<br>Section 5.1, 5.2<br>End of Midterm 2 Content |  |
| 11 | 30-Mar-26                                                      |  | 31-Mar-26                |  | 01-Apr-26                                                                    |  | 02-Apr-26                |  | 03-Apr-26                                                                             |  |
|    | Diagonalization<br>Sections 5.3<br>Online HW10, Written HW8    |  |                          |  | Review Day<br>No Lecture<br>Exam 2 (Evening)                                 |  |                          |  | Diagonalization and Long-Term Behavior<br>Section 5.3                                 |  |
| 12 | 06-Apr-26                                                      |  | 07-Apr-26                |  | 08-Apr-26                                                                    |  | 09-Apr-26                |  | 10-Apr-26                                                                             |  |
|    | Diagonalization Activity<br>Section 5.3                        |  |                          |  | Inner Product, Length, and Orthogonality<br>Section 6.1                      |  |                          |  | Orthogonal Sets and Orthogonal Projection<br>Section 6.2, 6.3<br>Late Drop Ends       |  |
| 13 | 13-Apr-26                                                      |  | 14-Apr-26                |  | 15-Apr-26                                                                    |  | 16-Apr-26                |  | 17-Apr-26                                                                             |  |
|    | The Gram-Schmidt Process<br>Section 6.4<br>Online HW11         |  |                          |  | Orthogonality and Gram-Schmidt Activity<br>Sections 6.1 - 6.4<br>Written HW9 |  |                          |  | Least-Squares Problems<br>Section 6.5                                                 |  |
| 14 | 20-Apr-26                                                      |  | 21-Apr-26                |  | 22-Apr-26                                                                    |  | 23-Apr-26                |  | 24-Apr-26                                                                             |  |
|    | Least-Squares Problems<br>Sections 6.5<br>Online HW12          |  |                          |  | Diagonalization of Symmetric Matrices<br>Section 7.1<br>Written HW10         |  |                          |  | The Singular Value Decomposition<br>Section 7.4                                       |  |
| 15 | 27-Apr-26                                                      |  | 28-Apr-26                |  | 29-Apr-26                                                                    |  | 30-Apr-26                |  | 01-May-26                                                                             |  |
|    | The Singular Value Decomposition<br>Section 7.4<br>Online HW13 |  |                          |  | Course Review<br><br>Written HW11                                            |  |                          |  | Course Review                                                                         |  |